

ANGEL OZA

Information and Communication Technology

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SUMMARY

ICT undergraduate with experience in embedded systems and hardware prototyping, with hands-on project experience in microcontroller-based control, sensor integration, LED/display systems, and wireless communication and building practical projects. Have taken up leadership roles in student organizations, managing teams and organizing technical events. Comfortable with problem-solving, working in teams, and learning new tools as required

EDUCATION

MARWADI UNIVERSITY

Aug 2023 -
May 2027

Bachelor of Technology - Information and Communication Technology

- Coursework: Data Structures and Algorithms, Object-oriented Programming, Database Management Systems, Operating Systems, Computer Networks, and Software engineering
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PROJECTS

1. Night Sky Display

- Developed an interactive night sky model using an Arduino-controlled LED matrix.
- Implemented GPS and orientation-based tracking to render real-time constellation positions.
- Applied graph-based mapping techniques for accurate star visualization.
- Designed dynamic display patterns to illustrate mathematical concepts on the LED matrix.

2. Hand Gesture-Controlled Wheelchair

- Designed and developed an assistive mobility device for paralysis patients using gesture-based control with flex sensors.
- Programmed a microcontroller for real-time motor control, ensuring reliable signal processing and responsiveness.
- Focused on user-centric design by improving accuracy, comfort, and ease of use.

3. SurakshaLink

- Developed an offline emergency communication system using ESP32 and LoRa.
 - Implemented structured wireless communication with alert prioritization.
 - Integrated acknowledgment handling for reliable node-to-node messaging.
 - Designed a real-time dashboard for communication monitoring and logs.
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RESEARCH EXPERIENCE

Dhirubhai Ambani University (Ongoing)

- Designing a **Digital Twin** of a coal mine to simulate autonomous fleet coordination, **IoT-based sensor monitoring**, and battery-powered **mobile Wi-Fi infrastructure**.

- Modeling dynamic **loading–hauling truck coordination**, incorporating resource depletion, shifting loading points, and adaptive route planning under real-time operational constraints.
 - Building a **Python-based simulation** with animated visualizations and performance analytics to evaluate fleet synchronization, communication coverage, and operational efficiency.
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SKILLS

Programming & Software Development

- Languages: C++, Python, Java, Kotlin
- Web Technologies: ReactJS, HTML, CSS, JavaScript

Machine Learning (Supporting)

- Frameworks: PyTorch, TensorFlow, RDK OS
- Libraries: Pandas, NumPy, Matplotlib

Hardware & Embedded (Supporting)

- Computer Architecture, Digital Systems
 - Embedded Systems: Arduino, Raspberry Pi, ESP8266/ESP32, Zigbee, D-Robotics RDK X5
 - Circuit Design & Simulation: PCB Design, Logic Design
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LEADERSHIP & STEM ACTIVITIES

- **Chair - IEEE MEFGI Student Branch (2026):** Led student operations, coordinated technical events, and collaborated with industry professionals.
- **IEEE WIE MEFGI Student Branch - Web Master :** Led the digital initiatives of the IEEE WIE student branch as a core member, managing the social website and amplifying IEEE WIEs outreach.
- **Convenor - Circuitology Club (2024 - '25):** Built a 110+ member technical community, secured USD 3,500+ in funding, and organized 30+ technical sessions and STEM outreach programs.
- **Marketing Lead - Google Developer Group On-Campus MU:** As Marketing Lead at GDG on Campus, strategized and executed the chapters branding and communication initiatives, leading design, content, and outreach efforts to build a strong tech-driven community on campus.